

Exploring Deep Learning Approaches for Mammography Diagnostics with Down-Sampled and Cropped Images

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CERTIFICATE

This is to certify that the research work embodied in the dissertation entitled “**Exploring Deep Learning Approaches for Mammography Diagnostics with Down-Sampled and Cropped Images**” has been carried out at the School of Computational and Integrative Sciences, Jawaharlal Nehru University, New Delhi for the partial fulfilment of the requirement for the award of the degree of Master of Science in Computational and Integrative Sciences.

The dissertation embodies research work carried out by **Ankesh Kumar Pandey**, under the guidance of **Prof. Shandar Ahmad**, and has not been submitted in part for the award of any other degree or diploma to the best of our knowledge.

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My parents have been my greatest source of strength; this work is dedicated to them. I chose to work with an Indian-origin mammography dataset because my mother had breast cancer (CA Breast) and was diagnosed early, thanks to the non-invasive technique of Mammography. This personal experience inspired me to contribute to the field of cancer diagnostics in the way I could.

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Abstract:

This project was started in the III semester of my coursework and it is being continued in this semester as well. It investigates the impact of reducing resolution and cropping mammography images on the performance of deep learning models for breast cancer classification.

It is being hypothesized that lower resolution images can minimize radiation exposure for patients while maintaining diagnostic accuracy. Various popular deep learning models will be evaluated to determine the minimum effective resolution for accurate classification of mammograms as cancerous or non-cancerous. Advanced techniques such as transfer learning and data augmentation will be leveraged to enhance model performance, addressing challenges related to limited dataset and variability in image quality.

Following the identification of the optimal model and resolution, super-resolution techniques will be employed to up sample the classified images. This step aims to improve visual interpretation for clinicians, enabling better diagnostic decisions without compromising patient safety. The research is expected to enhance the efficacy of mammography interpretation and contribute to the ongoing discourse on balancing diagnostic accuracy with patient care in radiology. The findings are anticipated to inform future practices in mammographic imaging and deep learning applications in medical diagnostics.

Keywords: Deep learning, medical diagnostics, convolutional neural networks, medical imaging, healthcare, artificial intelligence.

Introduction:

(Surmacki *et al.*, 2013) Mammography is widely regarded as the gold standard for breast cancer detection (Surmacki *et al.*, 2013) employing X-rays to produce detailed images of breast tissue. This screening method is crucial for identifying advanced as well as early-stage cancers, which can significantly improve treatment outcomes and reduce mortality rates.

The American Cancer Society recommends annual mammography screenings for women aged 40 and older, emphasizing the importance of early detection in combating breast cancer. However, while mammography has proven to be an effective diagnostic tool, it is not without its drawbacks.

One of the primary concerns associated with mammography is the exposure to ionizing radiation. Each mammogram exposes patients to approximately 0.4 mSv of radiation, a dose equivalent to about 67 days of natural background radiation. For individuals undergoing annual screenings, this radiation exposure can accumulate over time, raising concerns about potential long-term health risks, including an increased likelihood of developing radiation-induced cancers. This issue is particularly pertinent for women who may undergo multiple screenings throughout their lives, highlighting the need for strategies that can mitigate radiation exposure without compromising diagnostic accuracy.

In light of these concerns, this project aims to explore innovative approaches that leverage deep learning techniques on low-resolution mammography images generated with reduced X-ray exposure. The hypothesis proposes that it is possible to accurately predict breast cancer using these lower resolution images, thereby significantly diminishing the radiation burden on patients. By harnessing artificial neural networks, this research seeks to bridge the gap between effective cancer detection and patient safety.

The project encompasses these primary objectives-

First, an investigation and review of various image downsampling techniques will be conducted to assess their applicability in mammogram classification. This will involve analysing how different downsampling methods affect image quality and classification performance.

Second, a robust classifier model will be developed to differentiate between diseased and healthy tissue using the downsampled data. This model aims to maintain high levels of

accuracy despite the reduced resolution of the input images. Lastly, transfer learning of the model will be tested as well.

Through this research initiative, it is anticipated that advancements in imaging technology and artificial intelligence can enhance breast cancer screening protocols while prioritizing patient safety. By reducing unnecessary radiation exposure without sacrificing diagnostic accuracy, this project could contribute significantly to the ongoing discourse on improving mammographic practices and ultimately lead to better health outcomes for patients undergoing breast cancer screening.

Background:

Current mammography practices are primarily optimized for manual diagnosis, relying heavily on the expertise of radiologists to interpret the images produced. While traditional mammography systems have made significant advancements in image quality and dose optimization, they still face inherent limitations in detecting subtle details that may indicate the presence of breast cancer. The close similarity in radiation attenuation coefficients among various breast tissues can make it challenging for radiologists to distinguish between normal and pathological findings, particularly when the features of interest are on the order of micrometers. Consequently, the accuracy of manual interpretations can vary significantly.

In contrast, deep learning models have shown great promise in enhancing diagnostic accuracy through automated image analysis. These models can process vast amounts of imaging data and identify intricate patterns that may remain unnoticed by the human eye. By leveraging convolutional neural networks (CNNs) and other advanced algorithms, deep learning approaches can extract relevant features from mammographic images more effectively than traditional methods. This capability not only enhances the detection of microcalcifications and other critical indicators of breast cancer but also allows for the analysis of lower resolution images without a significant loss in diagnostic performance.

The integration of deep learning techniques into mammography could revolutionize breast cancer detection by providing a more consistent and objective assessment of imaging data. By utilizing downsampled images generated with reduced X-ray exposure, this project aims to

explore how these advanced computational methods can maintain or even improve classification accuracy while minimizing radiation risks associated with traditional mammography. This research is particularly relevant in light of recommendations from organizations such as the American Cancer Society, which advocate for regular screenings but also acknowledge the potential health risks associated with repeated X-ray exposure.

Ultimately, by harnessing the strengths of deep learning models, this project seeks to contribute to a safer and more effective approach to breast cancer screening, ensuring that patients receive accurate diagnoses while reducing their overall radiation burden.

Review of Literature:

The integration of deep learning (DL) techniques into mammography represents a significant advancement in the field of breast cancer detection and diagnosis. Traditional mammography has long been the gold standard for screening, but it is primarily optimized for manual interpretation by radiologists. This reliance on human expertise can lead to variability in diagnostic accuracy, as subtle lesions and microcalcifications may be overlooked due to the limitations of human perception. Studies have shown that deep learning models can outperform human observers in detecting these critical details, providing a promising alternative to conventional methods.

Why are low resolution images necessary?

In X-ray imaging, as the pixel count increases (for example, when reducing pixel size to enhance spatial resolution), the area that each pixel covers decreases. Consequently, to maintain a consistent signal-to-noise ratio (SNR) and image quality, a greater total number of incident photons is required. This increase can indeed be approximated as growing with the cube of the pixel count because each pixel must receive a sufficient number of photons to accurately represent the image data. Specifically, if we halve the dimensions of each pixel (thus reducing the area by a factor of four), it would need to increase the total number of incident photons to achieve similar imaging performance (Grodzins, 1983).

On the other hand, the radiation dose is related to the total number of photons emitted during an X-ray exposure. When considering how dose relates to pixel count, if the pixel size is reduced (and thus increase pixel count), it may be needed to increase the total radiation output to ensure that each smaller pixel receives an adequate number of photons for accurate detection. This relationship can lead to a situation where decreasing pixel size results in a substantial increase in radiation dose—potentially increasing with the fourth power due to factors like increased exposure time and higher photon flux requirements (Grodzins, 1983). Mammographic radiation exposure does affect patients undergoing mammography, it poses potential health risks, particularly the risk of radiation-induced carcinogenesis. Screening mammography is effective in detecting malignant diseases early, the use of ionizing radiation can be associated with a risk of radiation-induced breast cancer. This risk is particularly relevant given that mammography involves exposure to low-dose X-rays, which can contribute to the overall radiation burden on patients. It's important to note that the radiation dose is balanced as part of a trade-off with image quality (Pauwels, Foray and Bourguignon, 2016).

Application of Deep Learning Techniques in Mammography Image Classification

Mammography image categorization has been greatly impacted by recent developments in deep learning, which have provided novel strategies and architectures to improve diagnostic precision. A 2020 study by Salama et al (Salama and Aly, 2021) presented a sophisticated paradigm that combines InceptionV3, DenseNet121, ResNet50, VGG16, and MobileNetV2 are several deep learning models that are used to categorize mammography images from datasets like MIAS, DDSM, and CBIS-DDSM into benign and malignant groups. By utilizing both Cranio Caudal (CC) and Mediolateral Oblique (MLO) views for increased diagnostic precision, this framework also included a modified U-Net model for segmenting breast regions in the mammograms, with the goal of helping radiologists discover breast cancer early.

The study effectively addressed the challenge of limited labeled data through transfer learning and data augmentation, achieving an impressive accuracy of 98.87% on the DDSM dataset.

In 2023, another study shifted focus to improving breast cancer risk prediction in younger women by developing deep learning methods to estimate mammographic density using low-dose mammograms taken at about one-tenth of the standard radiation dose. This research

evaluated the quality and reliability of density scores derived from low-dose images and analyzed the impact of image resolution and training conditions on prediction performance.

The researchers trained and tested their models on over 15,000 standard and simulated low-dose mammograms with known labels using two feature extraction techniques and an end-to-end training approach. They also evaluated performance using 296 corresponding standard and real low-dose images. The results indicated comparable prediction quality for both standard and low-dose images across all training conditions and resolutions. Notably, higher image resolutions improved performance for feature extraction methods, while the trained models consistently performed well across different resolutions.

(Squires *et al.*, 2024) Furthermore, the study revealed a strong correlation between predictions for standard and low-dose images, with Spearman rank correlation coefficients of 0.951 at low resolution and 0.956 at high resolution, highlighting the reliability of the approach (Squires *et al.*, 2024).

(Hamidinekoo *et al.*, 2018) A 2017 paper proposes a computer-based approach for breast cancer modelling known as the Mammography-Histology-Phenotype-Linking-Model. This model aims to create a mapping of features and phenotypes that connect mammographic abnormalities with their histopathological representations. The study discusses various challenges associated with this approach and highlights its potential contributions to clinical decision-making and treatment management. By linking imaging data with histological findings, the model seeks to enhance diagnostic accuracy and improve patient outcomes in breast cancer care. Prior research has employed deep learning techniques to classify mammograms as either healthy or diseased, as well as to categorize them into various BI-RADS classifications. However, only a limited number of studies have investigated the optimal image size for achieving the best disease detection outcomes (Hamidinekoo *et al.*, 2018). Researchers around the world from various domains have been doing experiments.

Use of super-resolution techniques:

Super-resolution is a powerful technique that creates high-resolution images from low-resolution ones, enhancing image quality without needing to upgrade existing imaging

hardware. This approach has significant potential in medical imaging. A study by Umehara K. et al. (Umehara, Ota and Ishida, 2017) explored this concept using a Super-Resolution Convolutional Neural Network (SRCNN) to improve the resolution of digital mammograms. The SRCNN was trained to reconstruct high-resolution images from low-resolution inputs, and its performance was compared to traditional linear interpolation methods, such as nearest neighbor and bilinear interpolation.

The study examined how well SRCNN-enhanced images correlated with clinical features of mammographic lesions, including breast density, BI-RADS assessments, and pathology-confirmed diagnoses. Image quality was assessed quantitatively using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). The findings revealed that SRCNN significantly outperformed linear interpolation methods in image quality ($p < 0.001$). Notably, SRCNN-enhanced images showed greater quality improvements in cases of dense breasts, high-risk BI-RADS categories, and pathology-confirmed malignant lesions compared to less dense, lower-risk, or benign cases ($p < 0.01$). The study concluded that SRCNN is an effective tool for enhancing digital mammograms, particularly for magnified images in more complex scenarios.

Breast cancer remains the leading cause of cancer-related deaths among women globally, according to the World Cancer Research Fund. Early detection significantly improves survival rates, making high-quality mammography vital for effective screening. However, achieving higher image resolution often requires increased radiation exposure, which poses health risks. To address this challenge, researchers have developed methods to enhance mammogram resolution without raising radiation doses.

Recent advancements in computer vision have focused on improving image resolution in areas like facial imaging. Inspired by these developments, Zheng J. et al. (Zheng, Fuentes and Leung, no date) adapted such techniques for medical imaging, particularly mammograms. Their study introduced an end-to-end method that generates high-resolution mammograms from low-resolution inputs created with minimal X-ray exposure. This approach uses a mesh warping algorithm for automatic image alignment, followed by a two-step super-resolution process

combining a PCA-based global model with a patch-based local model. Experimental results demonstrated that this method effectively produces high-quality mammograms from low-resolution images without requiring manual adjustments, offering a promising solution for safer and more efficient breast cancer screening.

Deep learning approaches are better in capturing minute details of low resolution images:

Current radiological practices often prioritize images that are visually interpretable by clinicians, potentially overlooking the superior pattern recognition capabilities of machines. This approach can result in unnecessarily high-dimensional images and increased radiation exposure, which has not been adequately addressed in image analysis studies. In a study conducted by Vinayak N. et al (Vinayak, Pandey and Ahmad, 2024), they have concluded that a CNN model like EfficientNetB0 is very promising in predicting diseased versus healthy condition by using low resolution diagnostic images like chest X-ray image. In this study the researchers tried to do a classification between COVID-19, Pneumonia and healthy individuals by looking at their chest X-ray images using CNN models. Existing research on detecting COVID-19 and pneumonia using chest X-ray images typically involves reducing the original image size to various lower dimensions, often around 224 x 224 pixels. This study aims to investigate the impact of image size on model performance by examining sizes ranging from 300 x 300 pixels down to a minimal 1 x 1 pixel. The specific sizes analysed include: 1 x 1, 2 x 2, 4 x 4, 8 x 8, 16 x 16, 32 x 32, 64 x 64, 128 x 128, 224 x 224, 256 x 256, and 300 x 300 pixels.

The findings indicate that as image size decreases, the visible information to the human eye also diminishes; however, neural networks can still effectively process lower-dimensional images. This raises two key questions: first, which X-ray dimensions are optimal for effective disease detection? Second, if lower-dimensional images can accurately identify diseases, can X-rays be collected at these reduced dimensions to minimize patient radiation exposure? The study emphasizes the trade-off between radiation exposure and detection performance across different image sizes. The study investigates three binary classification problems: Covid-19 vs. Normal, Covid-19 vs. Pneumonia, and Pneumonia vs. Normal. The findings indicate that there is negligible learning from image sizes of 1 x 1 to 4 x 4 pixels, with noticeable learning

beginning at 8 x 8 pixels and continuing to improve up to 128 x 128 pixels. Beyond this size, performance gains are minimal, particularly from 128 x 128 to 300 x 300 pixels.

For the Covid-19 vs. Normal classification, the area under the curve (AUC) for test data at 128 x 128 pixels is 99.94%, with only slight increases at larger sizes (99.97% for 224 x 224 pixels, 99.98% for 256 x 256 pixels, and reaching 100% at 300 x 300 pixels). At a lower size of 64 x 64 pixels, the AUC score is significantly lower at 95.35%.

In the Covid-19 vs. Pneumonia task, the AUC score at 128 x 128 pixels is 99.54%, increasing slightly with larger sizes (99.94% for 224 x 224 pixels, 99.98% for 256 x 256 pixels, and dropping slightly to 99.97% at 300 x 300 pixels). For the Pneumonia vs. Normal classification, the AUC scores are lower overall, with a score of 97.99% at 128 x 128 pixels and increases to 98.03%, 98.8%, and 99.05% at the larger sizes of 224 x 224, 256 x 256, and 300 x 300 pixels, respectively. They have shown that, while model performance shows only slight improvements when increasing image size from 128 x 128 pixels onward, the radiation dosage for patients increases significantly. Specifically, achieving a small performance gain requires approximately nine times more radiation exposure at 128 x 128 pixels, and increasing the image size to 300 x 300 pixels raises the dosage by around 30 times. The dataset contains actual image sizes ranging from approximately 800 x 800 pixels to 2000 x 2000 pixels, with an average size of about 1000 x 1000 pixels. At this larger size, the radiation dosage is 3,725 times greater than that for a 128 x 128 pixel image and about 400 times more than for a 224 x 224 pixel image.

This study demonstrates that deep learning models can effectively learn from low-dimensional chest X-ray images, where human interpretation may fall short. By leveraging advancements in artificial intelligence, it is possible to capture smaller X-ray images, significantly reducing radiation exposure for patients. To create higher-dimensional images for medical practitioners, generative models such as Generative Adversarial Networks (GANs) can be used. This approach benefits both patients and doctors by minimizing radiation exposure while providing high-quality scans for review.

The findings suggest that the size of X-ray images and associated radiation exposure can be reduced, allowing for informed decisions regarding the exposure-performance trade-off on a case-by-case basis. Additionally, the general design of this study indicates that its methods could be applied to other medical diagnostic techniques involving X-rays or other forms of high-intensity radiation.

This study has been of crucial importance in this mini project, as it provides a solid foundation for such kind of low resolution diagnostics research and makes a strong argument in the favour of use of deep learning models for diseased versus healthy classification based on radiological images, although radiological images have been the images of choice, other images such as whole slide images can be used as well.

Feature extraction-based approach for classification:

The transformer architecture, introduced in the revolutionary paper "Attention is All You Need" by Vaswani *et al.* in 2017 (Vaswani et al., 2017), marked a significant departure from recurrent-based neural networks for sequence modelling tasks. The key innovation of transformers lies in their self-attention mechanism, which allows the model to weigh the importance of different elements in a sequence when making predictions.

The self-attention mechanism can be described as mapping a query and a set of key-value pairs to an output. As explained in the original paper, "The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a function of the query with the corresponding key." (Vaswani et al., 2017) This enables the model to focus on relevant parts of the input sequence regardless of their position.

(Dosovitskiy *et al.*, 2020)The Vision Transformer (ViT) adapts this architecture to image analysis by treating an image as a sequence of patches. As described by (Dosovitskiy *et al.*, 2020), the ViT model "applied directly to sequences of image patches can perform very well on image classification tasks." The process involves:

1. Splitting an image into fixed-size patches
2. Linearly embedding each patch
3. Adding position embeddings
4. Feeding the resulting sequence of vectors to a standard Transformer encoder
5. Using a classification token (CLS) to obtain the final representation for classification

This approach allows the model to learn global relationships between different regions of the image without the inductive biases inherent in CNNs.

All of this literature has been studied and a prominent research gap was found out which was that although deep learning techniques have and are being applied in image based diagnosis of breast cancer, it hasn't considered the effect of low resolution images and the use of transformers for extracting latent space features.

Objectives:

Three objectives of this project have been defined; these are-

1. To investigate and review image down sampling techniques applied to mammogram classification.
2. To develop a classifier model which classifies the down sampled data into diseased vs healthy.
3. To attempt to build a model which is able to achieve transfer learning, and predicts across datasets.

The first objective encompasses majorly two areas, researching about the appropriate dataset and downloading it and then preprocessing the images. Preprocessing includes converting the images to workable file format to start with. Medical images are stored in DICOM (.dcm) file format as a standard, so converting these images to a lossless file format (like .png). Studying various different image downsampling and cropping techniques and applying them to the dataset, checking the redundancy in the dataset, compiling and cleaning the image metadata are some of the preprocessing which needs to be done.

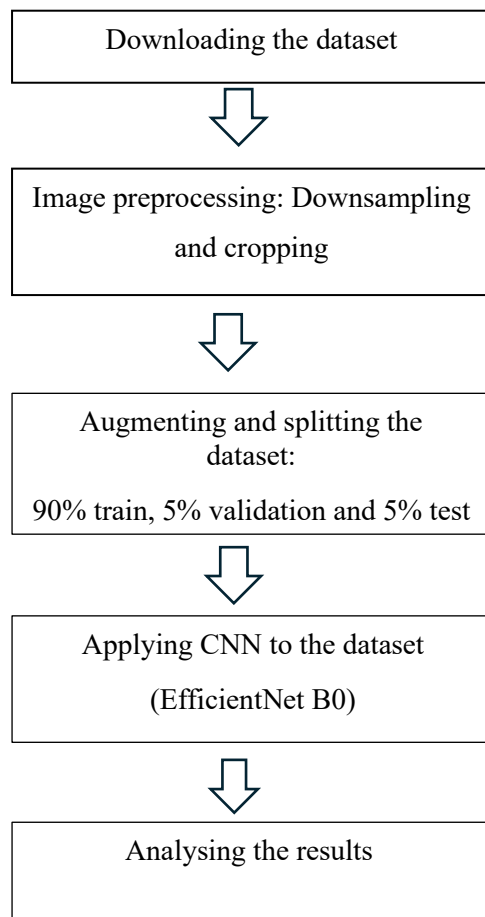
The second objective includes exploring and studying various neural networks, particularly convolutional neural networks (CNNs), as these are the neural networks specifically designed and used for image classification. The neural networks once chosen to work with can be then applied on our dataset, at various different image resolutions, then their performance can be recorded to find the optimum image resolution which is good for predicting the disease on the one hand and is able to save the patients from unnecessary X-ray exposure as well.

The third objective tries to learn about whether the model which is being developed works across datasets or not? Whether the performance which is being recorded by training and testing on datasets of one demography works on datasets of other demography or not? These are the research questions which will be dealt under this objective.

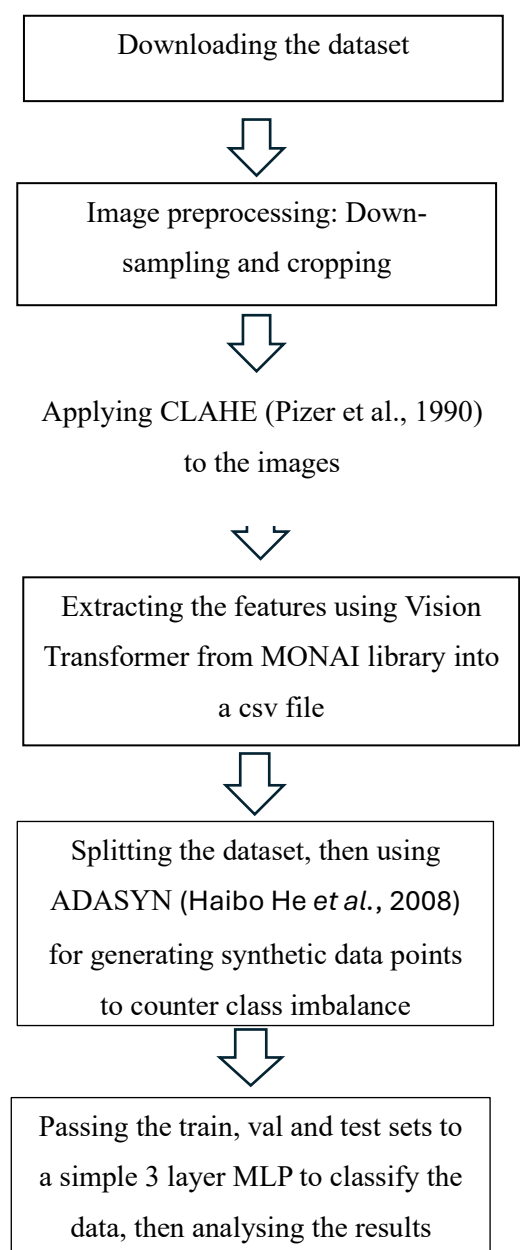
Methods

A comprehensive workplan was decided upon to start working on this project. Each part of the plan is being explained here in detail. A flowchart is being provided here as well.

1. Workflow for using the CNN model



2. Workflow for using the Vision Transformer approach



Datasets:

There are various mammography image datasets available, some of them are-

1. CBIS-DDSM (Curated Breast Imaging Subset of Digital Database for Screening Mammography) - The CBIS-DDSM is a refined and standardized version of the original DDSM, which was first created in 1997. Developed by the Cancer Imaging Archive and maintained under the Cancer Imaging Program of the National Cancer Institute (NIH, United States), this updated dataset was introduced in 2017. It contains 2,620 digitized film mammography studies, classified into normal, benign, and malignant cases, all supported by verified pathology information. This comprehensive dataset is an invaluable resource for developing and evaluating decision-support systems aimed at improving breast cancer diagnosis.
2. MIAS (Mammographic Image Analysis Society) Dataset - Established in 1992 by a consortium of UK academic institutions, the MIAS dataset comprises 322 scanned film mammograms, archived in Portable Gray Map (PGM) format. The dataset includes a diverse range of cases classified as normal, benign, or malignant, along with verified pathology information that serves as ground truth for research and algorithm development. One of the primary motivations behind creating the MIAS dataset was to facilitate advancements in computer vision and improve the ability of computers to interpret medical images.
3. INbreast Dataset - It is a publicly available repository of full-field digital mammograms, released by the Hospital de São João in Porto, Portugal, in 2011. Comprising 410 high-resolution digital mammographic images in DICOM format, the dataset includes a diverse range of cases classified into four distinct categories: masses, calcifications, asymmetries, and distortions. Each pathology within the dataset has been confirmed histologically through core biopsy or post-surgical specimen assessment, ensuring a high level of accuracy and reliability. Despite its strengths, the relatively small size of the INbreast dataset limits its use primarily to validation or supplementary purposes in larger studies.

4. MAMOS_1000000004 Dataset - It is a mammography images dataset of Indian population. It was generated at All India Institute of Medical Sciences, New Delhi under the project titled ‘An Opportunistic screening mammography dataset from a screening-naive population’. This dataset is hosted on the Indian Biological Images Archive website, a unit of Indian Biological Data Centre (IBDC). It is an open-source dataset containing 3577 images, in which 1869 Full Field Digital Mammography & 1708 Synthetic Mammography images are there. Images in this dataset are in DICOM (.dcm) format. It provides breast-level imaging data (BIRADS category and breast density) along with ground truth labels based on histopathology for cancers and follow-up scans for noncancers.

The “MAMOS_1000000004 Dataset” was chosen for this mini project due to 3 major factors-

1. This dataset has been created in India, thus making it more relevant for a study being conducted in this country.
2. All the other credible datasets such as those which are discussed previously, were generated fairly long ago, and the MAMOS_1000000004 Dataset was released in September 2024, making it a novel data to work with.
3. The size of this dataset is approximately 67 GB, making it much more manageable as compared to datasets like CBIS-DDSM, sized around 163 GB.

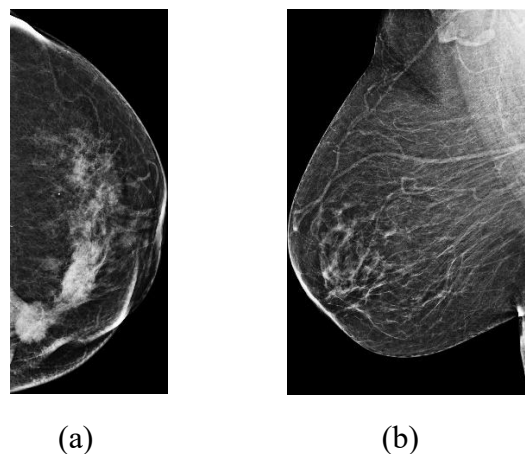


Fig 1: Samples from the dataset (a) Cancer (b) Not cancer

Image Preprocessing and Down-sampling:

After acquiring the dataset, it was processed to make it workable according to our needs. The preprocessing steps are described below.

1. Images were in DICOM format; DICOM stands for Digital Imaging and Communications in Medicine, a standard format for storing, transmitting, and displaying medical images and related data, it was developed by jointly by National Electrical Manufacturers Association (NEMA) and the American College of Radiology (ACR). DICOM is used as the format of choice for storing medical images mainly because of the metadata which it carries. It has around 300 metadata attributes such as patient information, study information, modality (e.g. CT, MRI, Mammography etc), image information (e.g. image position, orientation, pixel array, pixel spacing etc), equipment information, acquisition parameters (e.g. kilovoltage peak, exposure time, tube current etc).

While DICOM file format is great for data storage it is not that good for working with the data, as due to its metadata the individual file size becomes hefty, due to these constraints the images were converted in a lossless file format .png. This helped in reducing the file size by a factor of 10 while preserving the details of the image. After conversion the dataset's size was 6.7 GB.

2. Once all the images were converted, they were cropped to further reduce the redundancy in them, as the black region in a mammogram acquires 60-90% of the area it is imperative to remove the black region from the image as much as possible, while preserving all of the tissue area. It makes the computation much more efficient. For cropping many techniques were studied, namely-
 - 2.1.1. Intensity based thresholding- Very high efficiency, moderate accuracy
 - 2.1.2. Contour detection- High efficiency, moderate to good accuracy
 - 2.1.3. Convex hull- Moderate efficiency, high accuracy
 - 2.1.4. Deep learning segmentation- Moderate to low efficiency, very high accuracy.

Based on the advantages and the drawbacks offered by these methods, Convex Hull was found to be the optimum method for cropping the images. This algorithm finds the

smallest convex shape that encloses a set of points in a plane or space. In the context of breast tissue segmentation, it helps define the contour of the breast region and exclude surrounding non-tissue area by wrapping the breast tissue outline.

A python script was written to do the task of cropping the images, it used libraries like cv2 and NumPy. The output got through this method was exceptionally good.



Fig 2: Before cropping

After cropping

3. Indian Biological Images Archive provides other details of the dataset in formats like .xlsx (excel spreadsheet) and .csv (comma separated value file). Thus, the accompanying data was downloaded as three distinct .csv files, which contained sample id, experiment id, BIRADS categories, ground truth label and other features. In this stage of preprocessing the most important features of the dataset were extracted from these three files and compiled into one file. The final file contained sample id, image file name, and ground truth label. This file was further used as the file which contained the labels of all the images in the dataset. This was done with the help of python library pandas.

- **Samplewise normalisation:**

It was observed that the images were having widely varying pixel intensity distributions, which is a common trait in medical images. Sample wise normalisation of ImageDataGenerator functionality of Keras API was used to do the Z-score normalisation. The images were first rescaled to have a pixel intensity between 0 to 1 by dividing the pixel values by 255. Then Z-score normalisation was done on the rescaled images.

This helped the model in focussing on structural patterns rather than absolute intensity values. Unnormalized images may introduce biases due to different contrast levels,

brightness variations, or scanner-specific characteristics. Sample-wise normalization helps the model generalize better across different datasets and imaging conditions.

Deep Learning:

This has been the most crucial and resource intensive step. Various DL models were studied and it was decided that the deep learning step will start with the model EfficientNet B0. EfficientNet-B0 is the baseline model in the EfficientNet family of convolutional neural networks (CNNs), developed by Mingxing Tan and Quoc V. Le from Google Research in 2019(Tan and Le, 2019). This architecture strikes a balance between high accuracy and computational efficiency, making it ideal for applications ranging from mobile and edge computing to image processing tasks.

- Key Features of EfficientNet-B0:
 - Compound Scaling: EfficientNet-B0 uses a unique compound scaling method that scales the network's depth, width, and resolution uniformly based on fixed coefficients. This allows the model to adapt efficiently to varying computational resources while maintaining strong performance across tasks.
- Architectural Highlights:
 - MBConv Blocks: The backbone of EfficientNet-B0 consists of Mobile Inverted Bottleneck Convolution (MBConv) blocks. These blocks improve performance with depthwise separable convolutions and squeeze-and-excitation (SE) mechanisms, enhancing the network's ability to represent features effectively.
 - Layer Structure: The model has 237 layers organized into seven blocks, each configured to optimize learning and performance.
 - Parameters: For this specific implementation, the model contains 5,330,571 parameters, of which 5,288,000 are trainable and 42,571 are non-trainable.
- Performance:

EfficientNet-B0 achieves an impressive top-1 accuracy of 77.1% on the ImageNet dataset while using significantly fewer parameters compared to other models. This combination of accuracy and efficiency makes it particularly well-suited for deployment in environments with limited computational resources.

EfficientNet Architecture

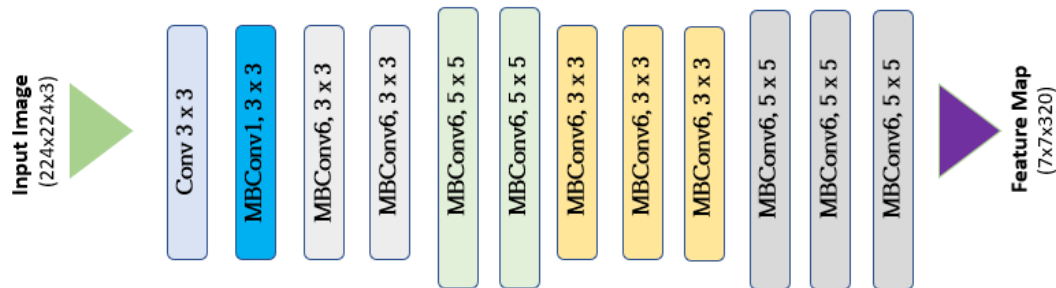


Fig 3: A schematic diagram of EfficientNet Architecture; Courtesy- WisdomML

Dealing with class imbalance in the data:

After cleaning the dataset, and the file containing all the labels for the images it found that there is severe class imbalance in the dataset. Here is a table summarizing the imbalance:

Cancer images	299
Non cancer images	3278

To deal with this imbalance in the data, data augmentation strategies were used for increasing the number of data points in the minority Malignant class.

Class weights were also defined to penalise the model for classifying the majority class wrongly and reward the model for correctly classifying minority class.

This discrepancy of the dataset has been a big hurdle while working with this data, in the progress report presented in the mini project report of III semester, it was mentioned that undersampling of the majority class was done to counter the class imbalance, in this semester a different approach of calculating and using the “Class weights” for training has been explored.

- Data augmentation:

The dataset was artificially expanded by using python code by the ImageDataGenerator from Keras API, through the application of various transformations to the images in the Malignant class.

- Augmentation Parameters -

1. Rotation range = 15: The image was randomly rotated within a range of -15 to +15 degrees, enabling the model to become invariant to small input rotations.
2. Width shift range = 0.1: Horizontal shifts of up to 10% of the image width were applied, simulating translations to increase robustness against slight shifts.
3. Height shift range = 0.1: Vertical shifts of up to 10% of the image height were applied, enhancing robustness against vertical translations.
4. Shear range = 0.15: A shearing transformation was applied, skewing the image along one axis, which is useful for simulating perspective distortions.
5. Zoom range = 0.11: Random zooming in or out by up to 11% was applied, making the model robust to variations in scale.
6. Horizontal flip = True: The image was flipped horizontally with a 50% probability, which is particularly useful for images where horizontal symmetry is meaningful (e.g., chest X-rays, mammograms).
7. Brightness = Brightness of the images was varied in a range of 0.7 to 1.3.
8. Fill mode = 'nearest': Missing areas caused by transformations such as rotation or shifting were filled using the nearest pixel values.

- Processing Images in minority class

The images in the Malignant class were first split into train, validation and test sets by specifying these directories. Each of these sets had two classes Benign and Malignant. The 'Malignant' images from the train directory were iterated through by the code. Each image was loaded and converted into a NumPy array for further processing. The array was reshaped to include a batch dimension $((1,) + \text{img_array.shape})$, as batch input is expected by the datagen.flow method.

- Generating Augmented Images

Augmented images were generated separately by the datagen.flow method, based on the specified augmentation parameters. The augmented images were then saved.

For every original image, 11 augmented images were generated. This ensured the augmented dataset was balanced and not overly dominated by artificially generated samples. Here is a table showing before and after statistics of the augmentation-

Training Images	Before augmentation	After augmentation
Malignant images	240	2640
Benign images	2621	2621
Total	2861	5261

No augmentation was applied on the validation and test set images to maintain the real life consistency in the images.

Validation Set	No. of Images
Malignant images	30
Benign images	328
Total	358

Test Set	No. of Images
Malignant images	30
Benign images	328
Total	358

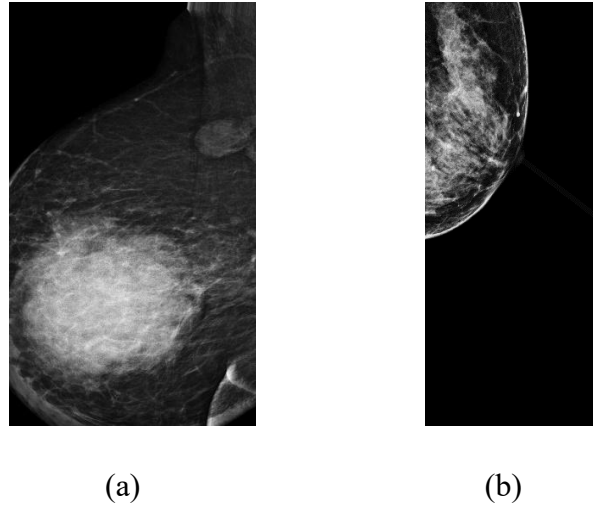


Fig 4: Augmented images: (a) zoomed (b) rotated, vertically shifted

Splitting the dataset and the Model run:

Once the class imbalance was taken care of by defining class weights the dataset was split in 90:5:5 ratio, i.e. 90% of the images in the training set, 5% images in the validation set and 5% images in the test set. EfficientNet B0 was run on the dataset for 5 different resolutions 300x300 px, 224x224 px, 175x175 px, 128x128 px and 64x64 px.

Early stopping criteria with a tolerance of 6 epochs was applied. ElasticNet regularisation (L1 and L2 regularisation) was applied with the L1, L2 values of 2.4396×10^{-5} and 2.9038×10^{-6} respectively. Train batch size was taken as 32 and validation batch size was taken as 16, to ensure better implementation of regularisation. A learning rate scheduler was used which monitored validation accuracy and reduced the learning rate by a factor of 0.5 upon continued stagnation of validation accuracy for 6 epochs.

Important statistics like training accuracy, training loss, validation accuracy, validation loss were recorded and plotted.

CNN Results:

The results which were obtained were betterment over the last reported work, as in the last report a test set wasn't included and many if the parameters were yet to be tuned like. Here in the work which is being reported a test set was included and all the tuned hyperparameters were utilised. Here is a table summarizing all the statistics which were collected during the neural network run.

Here are the accuracy and loss plots too-

Resolutions	Val. Accuracy	Val. Loss	Test Accuracy	Test Loss
300x300 pixels	0.92	0.08	0.92	0.09
224x224 pixels	0.87	0.19	0.83	0.17
175x175 pixels	0.93	0.12	0.92	0.15
128x128 pixels	0.91	0.11	0.91	0.10
64x64 pixels	0.86	0.15	0.85	0.13

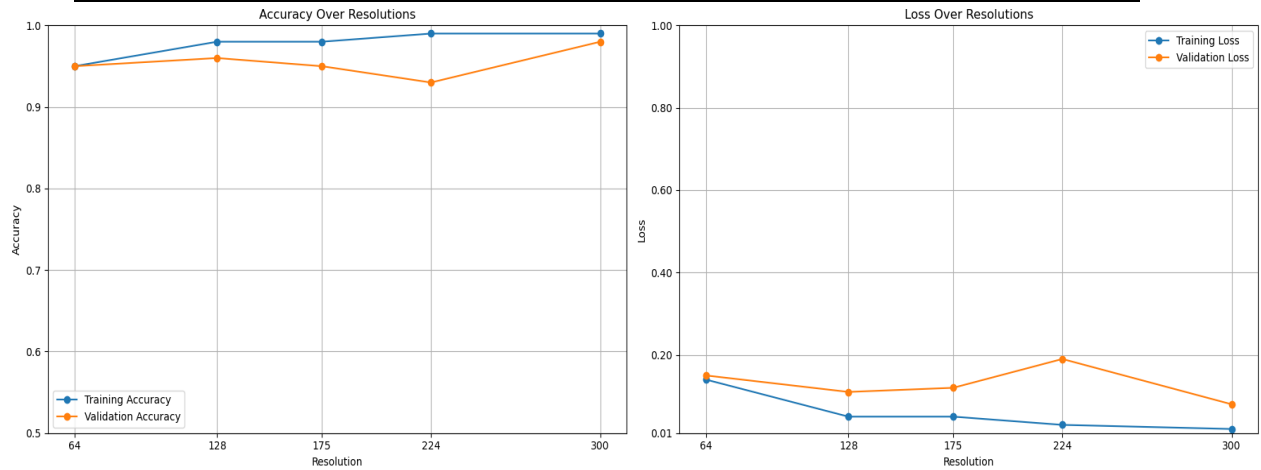


Fig 5: (a) Accuracy versus Resolutions

(b) Loss versus Resolutions

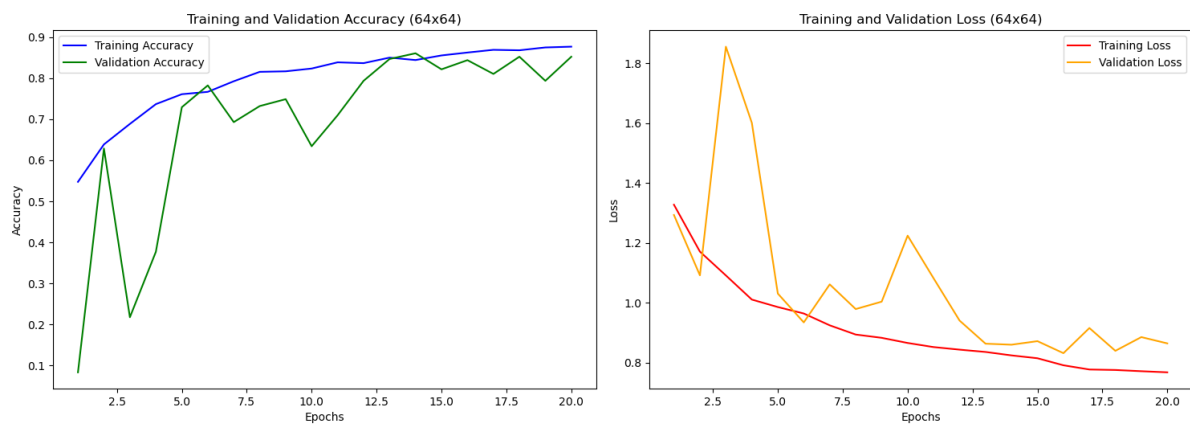


Fig 6: Training History for 64x64 pixels image resolution

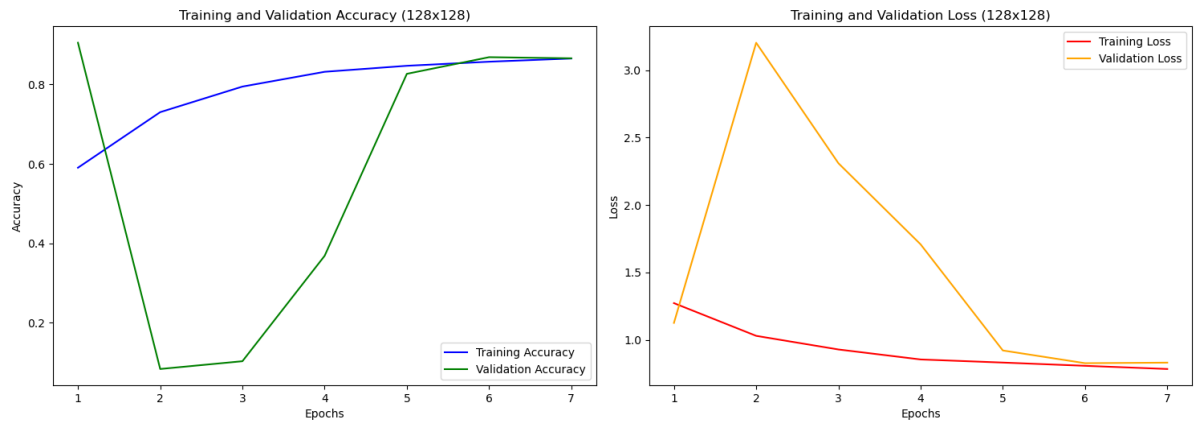


Fig 7: Training History for 128x128 pixels image resolution

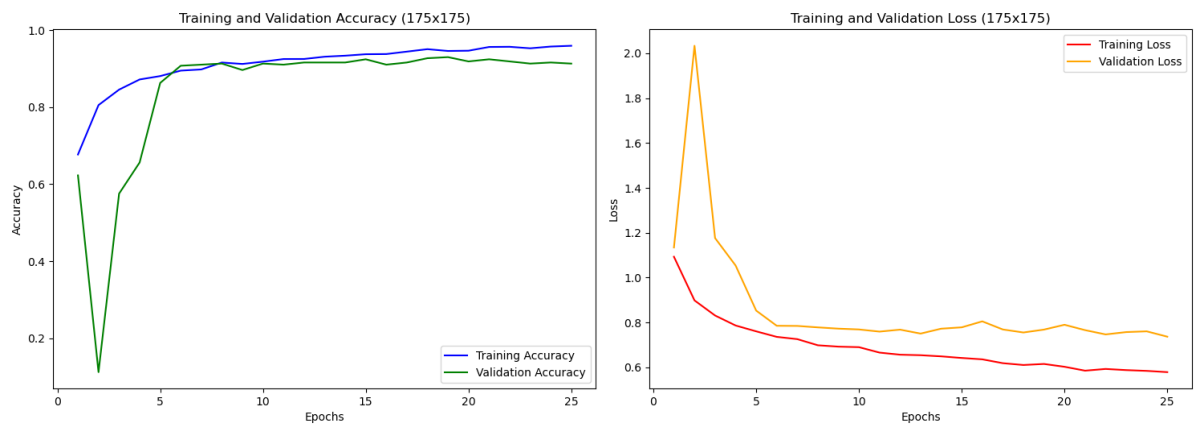


Fig 8: Training History for 175x175 pixels image resolution

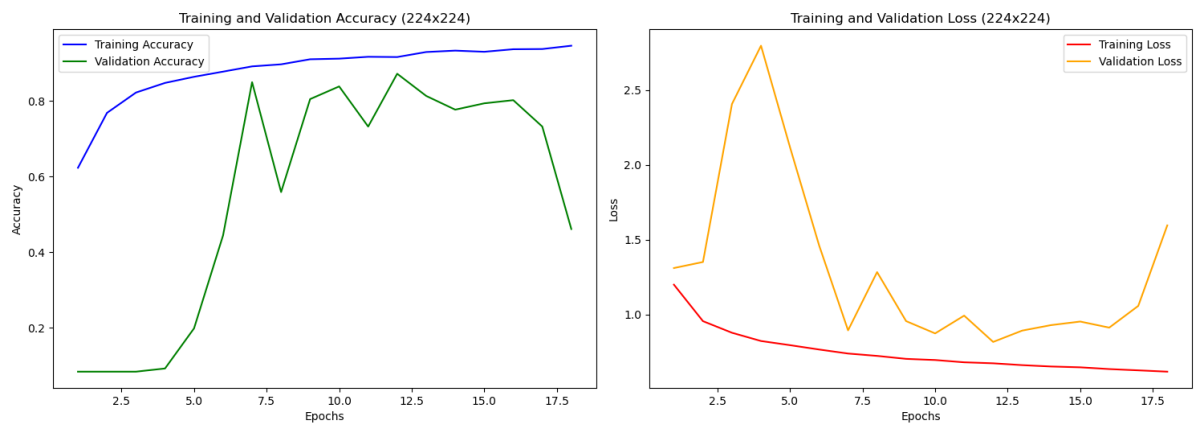


Fig 9: Training History for 224x224 pixels image resolution

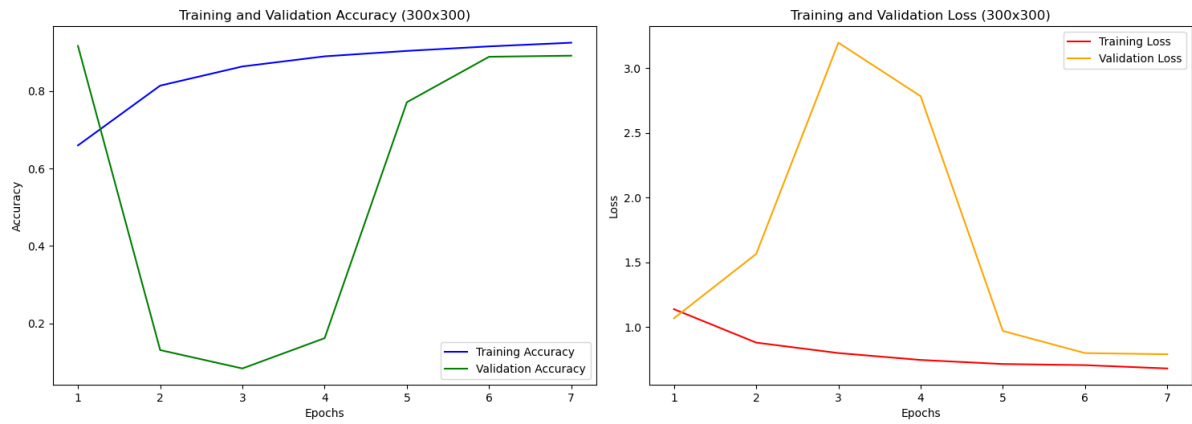


Fig 10: Training History for 300x300 pixels image resolution

Analysis and interpretation:

From the accuracy and loss plots we can see that as we move down from 300x300 pixels, we register a loss in accuracy, and gain in the loss of training and validation sets both, especially with the validation set.

As far as the success of this particular CNN (EfficientNetB0) with our dataset and hyperparameters of choice is concerned we are concluding that an image of 175x175 pixels is good enough for EfficientNetB0 for classifying the Benign versus Malignant images with an accuracy of 0.92 (92%) while registering a loss of 0.15.

Transformer based feature extraction:

For this study, we implemented a ViT-based approach for mammogram classification using the MONAI library and the PyTorch framework. MONAI stands for “Medical Open Network for AI” originally developed by NVIDIA and King's College London. It is a collaborative initiative that brings together leading researchers, clinicians, and industry experts to advance medical imaging AI. The implementation follows these key steps:

Data preparation and preprocessing:

Our mammography dataset contains 3,577 images, of which 3,277 are benign and 300 are malignant, representing a significant class imbalance, common in medical imaging datasets.

Preprocessing steps included:

- a) Converting the images from their native DICOM (.dcm) format to .png for easy computation.
- b) Image normalization to standardize pixel values
- c) Using Contrast Limited Adaptive Histogram Equalization.

- Contrast Limited Adaptive Histogram Equalization:

Contrast Limited Adaptive Histogram Equalization (CLAHE), introduced by Pizer et al. (Pizer *et al.*, no date), is a widely used image enhancement technique, particularly effective in the biomedical domain. CLAHE improves local contrast in images by applying histogram equalization to small regions (tiles) rather than the entire image, and limits the amplification of noise by clipping the histogram at a predefined threshold. This method is especially beneficial for medical images, such as X-rays and retinal scans, where subtle contrast differences are crucial for accurate diagnosis. CLAHE has demonstrated superior performance over traditional methods, enhancing fine details while preserving overall image quality

Vision Transformer Feature Extraction:

We utilized the pre-trained Vision Transformer model from MONAI, which is specifically optimized for medical imaging tasks.

- **Model Architecture:**

1. Input and Patch Embedding:

- Input channels: 1 (grayscale image)
- Image size: 224 x 224
- Patch size: 16 x 16.

- Patch embedding block:
 - Conv2d layer ($1 \rightarrow 768$) with kernel size 16 and stride 16 to embed patches
 - Dropout with $p=0.0$ (no dropout)

2) Transformer Encoder (12 layers total):

- Each layer is a TransformerBlock with the following components:

a) LayerNorm 1:

- Applies layer normalization before self-attention

b) Self-Attention Block (SABlock):

- Linear layer to project input to Q, K, V: $768 \rightarrow 2304$
- 12 attention heads
- Output projection: $\text{Linear}(768 \rightarrow 768)$
- Rearranging operations for attention computations
- Dropout for output and weights, both with $p=0.0$

c) LayerNorm 2:

- Applies layer normalization before the MLP block

d) MLP Block:

- Two Linear layers:
 - $\text{Linear}(768 \rightarrow 3072)$, followed by GELU activation
 - $\text{Linear}(3072 \rightarrow 768)$
- Dropout after each linear layer with Dropout probability (p)= 0.0

3) Final LayerNorm:

- Applies LayerNorm(768) to the output after all transformer blocks

Using this model architecture only the extraction of features was done, for classification separate classification models were used.

After extracting the features, they were saved in a .csv file. The feature file had 3577 rows, for the same number of images and 767 columns for extracted features plus 1 column for class labels, 0 for benign and 1 for malignant.

Once we had the feature and label dataset, it was decided to work on managing the class imbalance in the dataset.

- ADASYN - Adaptive Synthetic sampling approach:

ADASYN (Adaptive Synthetic Sampling) is an advanced oversampling technique designed to address class imbalance in datasets, particularly improving the learning of minority classes. Unlike Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples uniformly for the minority class, ADASYN adaptively focuses on harder-to-learn samples by generating more synthetic data in regions where the minority class is sparsely represented. This targeted approach helps classifiers better distinguish minority instances, especially in complex or overlapping regions, leading to improved recall and overall model performance. Studies have shown that ADASYN can outperform SMOTE and other traditional methods, especially in challenging, highly imbalanced scenarios (Salifu *et al.*, 2025) .

Here is a table showing class distribution before and after using ADASYN:

	BENIGN	MALIGNANT	TOTAL
Before	3277	300	3577
After	3277	2936	6213

After doing the data augmentation, we went forward for classifying the data with a simple multilayer perceptron.

Classification using MLP:

We employed a simple MLP for classification. The MLP had 3 hidden layers with 512, 256, 128 neurons respectively in each layer, each having ReLU activation. The output layer had 1 neuron with sigmoid activation. The total no. of trainable parameters were 558,081.

L1, L2 regularization was applied, with values: L1= 0.000002 and L2= 0.000141. Class weights and weighted accuracy were calculated. Training history, AUC-ROC, confusion matrix were plotted.

- MLP Results:

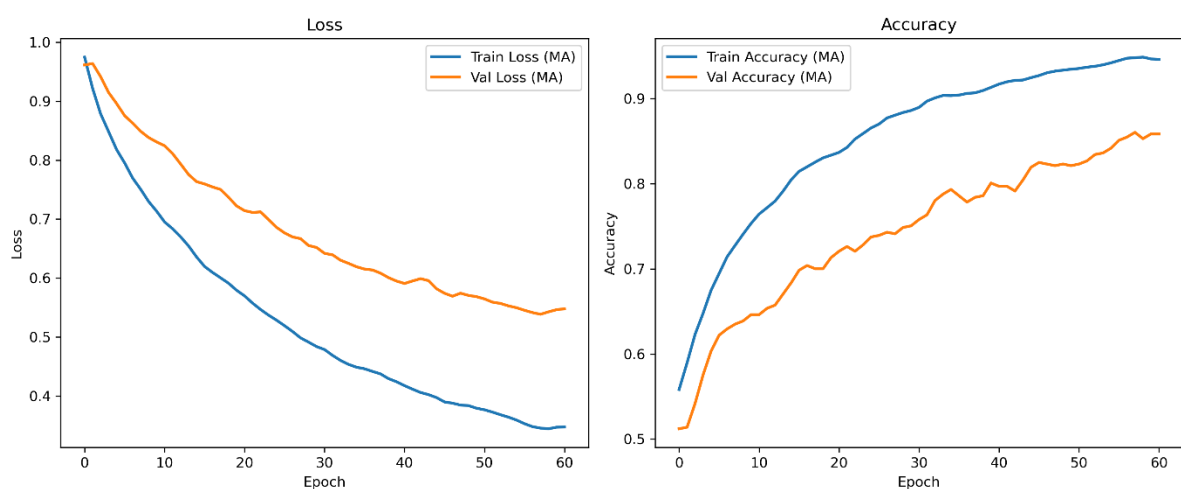


Fig 11: Training History

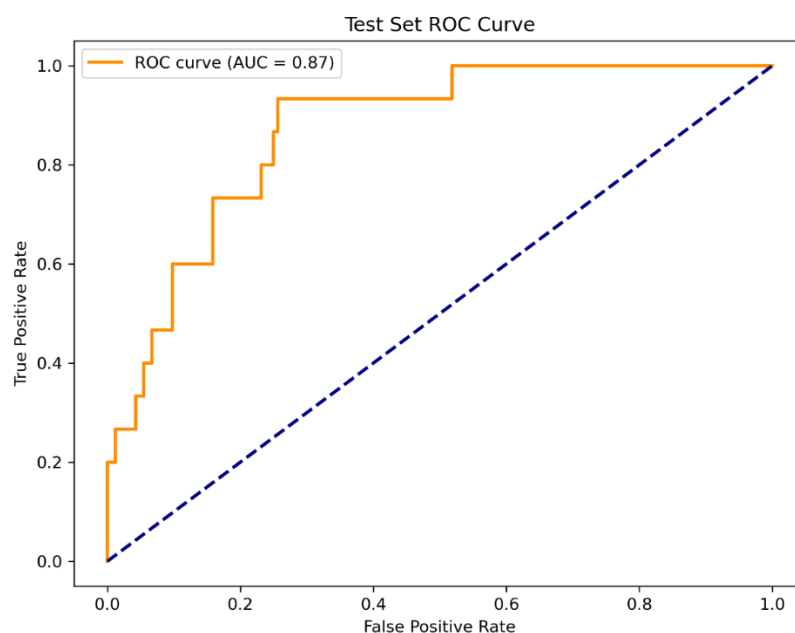


Fig 12: AUC-ROC on test set

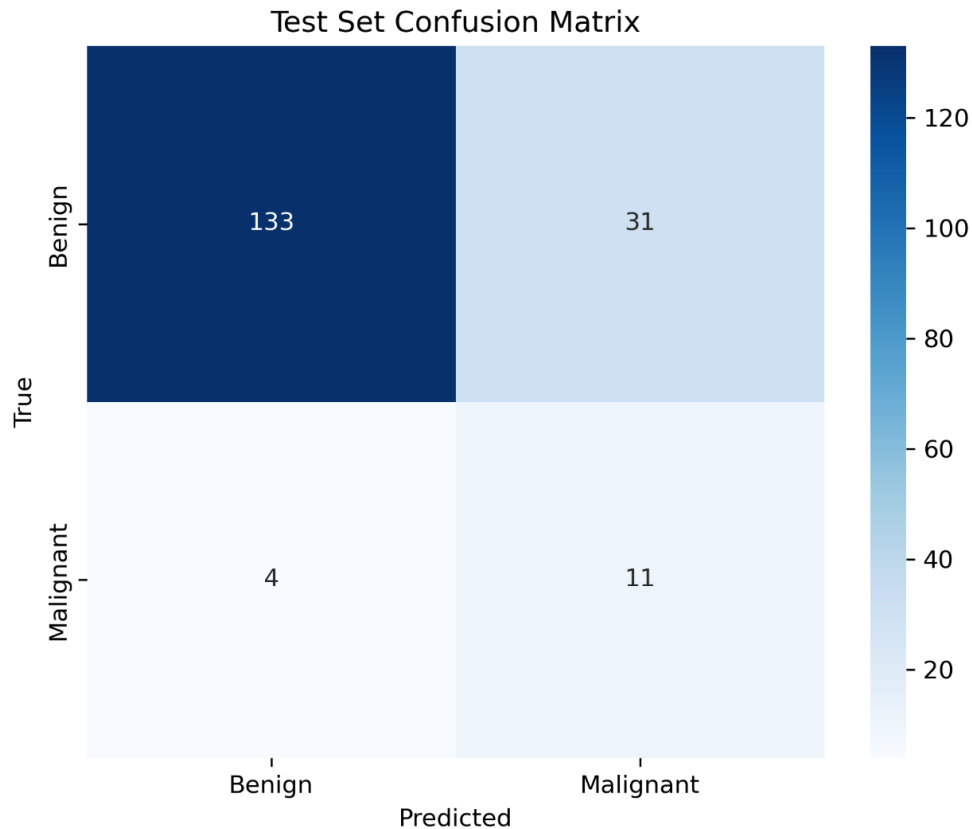


Fig 13: Test set confusion matrix at 0.5 cutoff

- Analysis:

After analysing the results from the work which was done in this semester after the mid semester evaluation, we can say that it is definitely an improvement over the past work which is done in this project. We are reporting an AUC-ROC of 87% on the test set i.e. on the unseen data our model is giving this result.

Conclusion:

In the tenure of this dissertation work, considerable work was done till now and new objectives were defined and fulfilled. It is in our consideration that the results which we have got can be refined further, using better hyperparameter tuning and running the models for larger number of epochs, we can extrapolate the results on a larger scale, which will be better for a real-life scenario. The third objective of this project, which was exploring and applying transfer learning still remains un fulfilled, we are optimistic that this objective will be completed as well.

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